

DOCUMENT 1.1 — BRAINSTORMING IDEA

Folder 1: Ideation Phase | ServiceNow Capstone Project Submission

Project Title: Hospital Appointment Management System

Application Scope: x_2065601_hospit_0

Platform: ServiceNow (Scoped Application)

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1. Problem Area Identification

Healthcare facilities, outpatient departments (OPDs), and private clinics frequently struggle with inefficient, manual appointment scheduling workflows. Traditional methods rely heavily on phone calls, physical logbooks, or disconnected software tools that lack synchronization with real-time clinical resources.

This operational bottleneck introduces significant systemic issues across clinical operations:

- **Double-Bookings & Scheduling Conflicts:** Manual logs fail to enforce concurrency controls, leading to overlapping patient appointments for a single doctor.
- **Low Visibility into Real-Time Availability:** Patients lack direct insight into active doctor shifts and available time slots, leading to high call volumes and long hold times at front desks.
- **Unmanaged Revenue Loss from Cancellations:** When a patient cancels an appointment, the vacated slot often remains unfulfilled due to the lack of an automated waitlist management mechanism.
- **Administrative Burden:** Front-desk personnel expend considerable time manually coordinating slot availability, managing daily queues, and verifying patient registration details.

2. Raw Brainstormed Ideas & Feature Evaluation

During the initial ideation phase, several potential features and subsystems were evaluated to address healthcare scheduling inefficiencies. The concepts were evaluated based on technical feasibility within ServiceNow, platform alignment, and value delivery.

1. Real-Time Atomic Slot Booking Engine Selected (In Scope)

A transactional booking mechanism that updates doctor slot availability and creates patient appointment records simultaneously within a single database transaction, preventing race conditions.

2. Event-Driven Auto-Promoting Waitlist System Selected (In Scope)

An automated waitlist queue that triggers immediately upon an appointment cancellation or no-show, auto-allocating the freed slot to the highest-priority waiting patient without manual intervention.

3. Multi-Persona Service Portal Dashboard Selected (In Scope)

A single-page, role-based AngularJS Service Portal widget displaying tailored operational interfaces for Patients, Doctors, and Front Desk/Admin users.

4. Automated Daily Doctor Queue & Lifecycle Tracker Selected (In Scope)

A real-time tracking interface for doctors and front desk staff to transition appointment states (`Booked` → `Checked-In` → `Completed` or `No-Show`).

5. Twilio-Based Automated SMS & WhatsApp Reminders Considered but Out of Scope

Integration with external telephony APIs (e.g., Twilio) to send automated SMS/WhatsApp appointment confirmations. Excluded to maintain focus on core ServiceNow scoped architecture, custom tables, and transactional integrity.

6. Telemedicine & Video Consultation Integration Considered but Out of Scope

Embedded WebRTC or Zoom API integration to support virtual medical consultations directly within the Service Portal. Deferred to future iterations due to external bandwidth requirements and third-party credential dependencies.

7. Real-Time Insurance Verification & Copay Billing Considered but Out of Scope

Direct integration with third-party payer clearinghouses to verify patient policy coverage and collect copays during booking. Excluded due to mock clearinghouse API complexities and regulatory compliance constraints.

3. Selected Core Idea & Justification

Selected Core Idea: A native ServiceNow Scoped Application (`x_2065601_hospit_0`) featuring a custom Service Portal (`hospital_booking` widget) that delivers transactional slot management, event-driven waitlist auto-promotion, and role-based lifecycle tracking for Patients, Doctors, and Front Desk staff.

Strategic Justification:

This core idea was selected because it directly solves the operational bottlenecks of hospital scheduling using native ServiceNow capabilities. By leveraging Scoped Application architecture, custom tables, business rules, script includes, and AngularJS Service Portal widgets, the solution ensures absolute transactional integrity (preventing double-bookings via atomic state updates) and minimizes administrative overhead through event-driven automation (auto-promoting waitlisted patients upon cancellation). This scope balances technical complexity with feasibility within the capstone timeline, delivering a fully operational, enterprise-ready workflow application without relying on external third-party dependencies.

4. Initial Feature List (Built Scope)

The final scope implemented in the application includes the following core technical features:

- **Custom Scoped Data Model:** Five dedicated tables in scope `x_2065601_hospit_0` (Patient, Doctor, Doctor Availability Slot, Appointment, and Waitlist) with strict relational integrity.
- **Service Portal Interface (`hospital_booking`):** A responsive, multi-tab AngularJS portal widget featuring custom views for Home, Patient Registration, Patient Booking & History, Doctor Registration, Doctor Queue & Slots, Front Desk/Admin, and Waitlist.

- **Atomic Slot Booking Engine:** Synchronous server-side processing that locks doctor slots (Available → Booked) and records appointments atomically to prevent race conditions.
- **Event-Driven Waitlist Auto-Promotion:** Business Rule logic that identifies cancellation events, evaluates waitlist priority order (priority_order), automatically reallocates freed slots, and updates waitlist states (Waiting → Promoted).
- **Role-Based Access Control (RBAC):** Custom Access Control Lists (ACLs) and ServiceNow roles enforcing data segregation across Patient, Doctor, and Front Desk personas.
- **Backend Verification Support:** Direct platform table mapping allowing system administrators to verify portal actions in native ServiceNow list views.